

Intelligent Detection System for Dirty Photovoltaic Panels based on YOLOv8

Shirui Liu¹, Chaoyue Lin¹, Zhe Xu^{1,2*}, Qiang Fu^{1,2*},

¹*School of Electrical Engineering and Automation, Tianjin University of Technology Binshui Xidao Extension 391, Tianjin, 300384, China*

²*Tianjin Key Laboratory of New Energy Power Conversion, Transmission and Intelligent Control
a1815755648@qq.com; linchaoyue@stud.tjut.edu.cn; fuqiang6369@hotmail.com;*

**Corresponding author: fuqiang6369@hotmail.com; xuzhe@live.cn*

Abstract - This article proposes an intelligent detection system for photovoltaic panel contamination based on YOLOv8n, named, which establishes a six-level classification standard to address the issue of quantifying contamination levels. The study uses a dataset of 3,862 photovoltaic panel contamination images, divided into training, validation, and test sets in a ratio of 7:2:1, and trains the model end-to-end using a YOLOv8n model. The network structure is optimized with the C2f module, resulting in a of 3.2M, which is 18% smaller than YOLOv5n, balancing accuracy and lightweight performance. Experiments were conducted under the Ubuntu20.04 environment, using a RTX 4060 GPU, and after 300 training cycles, the model achieved an mAP@0.5 of 0.832, with the highest AP value for the "Very Dirty" category (0.890) and the lowest for the "Clean" category (0.779), demonstrating the model's capability to detect different contamination levels. A human-machine interface based on PyQt has been developed, enabling real-time visualization of detection results and alarm linkage functions. This study provides an efficient solution for the automatic detection of photovoltaic panel dirt and the scheduling problem of cleaning robot, which helps to improve the operation efficiency and reliability of photovoltaic system.

Index Terms - YOLOv8; target detection; deep learning; photovoltaic panels; dirt recognition

I. INTRODUCTION

As the global energy structure accelerates its transition toward cleaner sources, photovoltaic power generation, as a crucial carrier of renewable energy [1-4], directly impacts the economic efficiency and stability of the energy system. However, issues such as dust accumulation on the surface of photovoltaic panels and bird droppings can reduce conversion efficiency by 5% to 30%, and may even cause hot spot effects leading to permanent damage to modules. Traditional manual inspection methods suffer from low efficiency, high costs, and strong subjectivity, while drones equipped with thermal imaging technology can achieve large-scale detection but struggle to quantify the degree of contamination.

In this context, intelligent visual inspection technology based on deep learning provides a new solution for monitoring the surface condition of photovoltaic panels. This paper proposes an intelligent recognition system for photovoltaic panel dirt based on the YOLOv8n model. By establishing multi-level dirt classification standards, it achieves real-time detection and graded evaluation of the surface contamination status of photovoltaic panels.

Regarding the qualitative and identification issues of dust accumulation on photovoltaic panels, scholars both

domestically and internationally have conducted extensive research. Pan Wuxing et al. found that the output power of photovoltaic modules decreases as the dust density increases [5]. If the panels are not cleaned for more than six months, power generation will decrease by 50%. Chang Zeyu et al. analyzed the impact of dust deposition on photovoltaic power generation [6]. Dust accumulation significantly suppresses the energy output of photovoltaic modules, with power generation decreasing exponentially with increasing dust concentration.

Zhao Bo et al proposed a method for detecting the degree of dust accumulation on solar panels [7], constructing a progressive predictive model that transforms electrical power loss and dust accumulation time. This method analyzes the degree of photovoltaic panels from the perspective of electrical power loss; however, in practical applications, different photovoltaic panels exhibit variations, with varying intrinsic electrical power losses, making this method less generalizable. Zhou Xiaoming established a relationship between reflected light intensity and the cleanliness of photovoltaic panels [8], analyzing the degree of dust accumulation through reflected light intensity. Finally, he designed and implemented a light reflection photovoltaic panel cleanliness detector, but this method has a high cost.

To address the aforementioned issues, this paper proposes an intelligent recognition system for photovoltaic panel dirt based on YOLOv8n. It can detect the degree of dirt and dust accumulation on photovoltaic panels in real time (six levels: clean; almost clean; dirty; completely dirty; dirty; slightly dirty; very dirty). This system aims to enhance the efficiency of photovoltaic cleaning robots while reducing equipment damage caused by panel contamination, thus extending their service life.

Additionally, as the latest lightweight model in the YOLO series, YOLOv8n significantly reduces parameters (only 3.2M) by simplifying network layers and introducing the C2f module (Cross Stage Partial Fusion), maintaining high precision. This improvement not only boosts the efficiency of photovoltaic panel pose detection but also provides better performance for the intelligent cleaning system, ensuring both the cleanliness of the panels and the longevity of the equipment. The final trained model supports export in ONNX format, is compatible with mainstream edge computing platforms such as NVIDIA Jetson and Huawei Ascend, and features a lightweight human-machine interaction interface developed based on PyQt, enabling visualization of detection results and alarm linkage.

II. STRUCTURE DESIGN OF INTELLIGENT IDENTIFICATION SYSTEM FOR PHOTOVOLTAIC PANEL DIRT

The whole process of system design is mainly divided into the following aspects.

1) Data processing: Collect the dirty photovoltaic panel image data set from the Internet and the laboratory, store it in the computer memory, then carry out photo screening and preprocessing, and finally use Roboflow to annotate the images.

2) Data division: The collected more than 3800 photos of photovoltaic panels were divided into training set, verification set and test set, and each data set was divided according to the ratio of 7:2:1.

3) YOLOv8 level detection: The divided data set is trained by YOLOv8. The training batch is selected for 300 rounds, and the best model is saved to best.pt for subsequent front-end interface and robot use.

4) Model evaluation: To comprehensively evaluate the model's recognition performance, several key performance metrics were adopted, including mean Average Precision (mAP), the calculation results of the loss function, and recall rate. These metrics collectively form a comprehensive evaluation system for model performance.

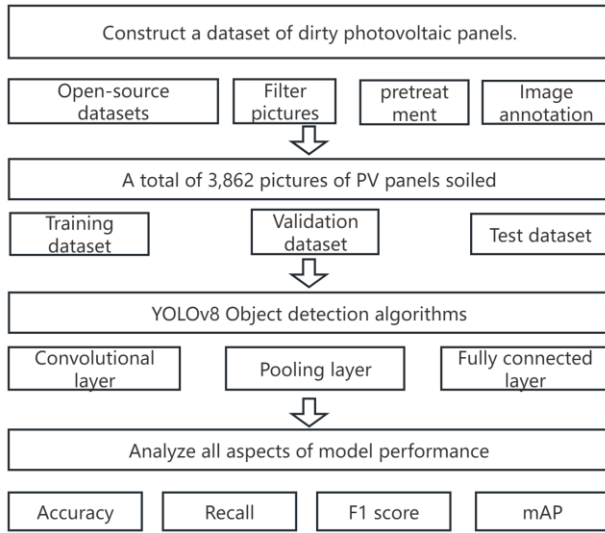


Fig. 1 Overall system flowchart

III. OVERVIEW AND EVALUATION OF YOLO MODEL

A. YOLO Model Structure

YOLO object detection algorithm is an end-to-end single-stage (One-Stage) object detection algorithm [9]. Its core lies in dividing the image into multiple regions and making predictions for these regions. YOLO divides the input image into several grids of 32x32 size, for example, a 416x416 image will be divided into 13x13 grids. When the center of the object is located within a certain grid, that grid takes responsibility Output the bounding box and category confidence of the object. Each grid has the capability to predict multiple bounding boxes and corresponding multiple

target categories, and the number of these bounding boxes and categories can be flexibly configured according to actual needs. The output of the YOLO algorithm is a feature map, which contains information about the bounding boxes and category confidence for each grid. This paper adopts the latest YOLOv8 model of YOLO [10], released at the end of 2022. It is the latest YOLO series model, featuring a new SOTA model and a novel network backbone structure, as shown in Figure 2.

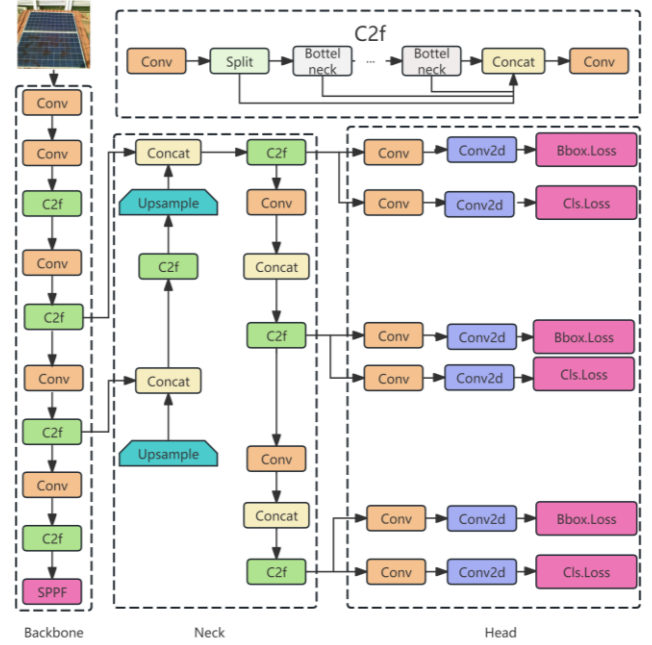


Fig. 2 YOLOv8 network architecture diagram

The entire network is divided into the Backbone part and the Head part. YOLOv8 inherits the essence of previous-generation networks, adopting the CSP design concept for its backbone and Neck sections [11]. It upgrades the C3 module in YOLOv5 to the more gradient-rich C2f module and removes the convolutional structure in the PAN-FPN upsampling stage of YOLOv5, directly feeding features output from different stages of the Backbone into the upsampling operation. The model provides different-sized versions at N/S/M/L/X scales to meet the needs of various industrial fields. This paper designs an intelligent soiling detection system for photovoltaic panels based on the YOLOv8 model. By configuring model parameters to train the classification of soiling levels on photovoltaic panels, the optimal model for deployment applications is obtained.

B. Loss Function

YOLOv8n, the loss function is CIOU, and the formula of the loss function is shown in formulas (1), (2), (3) and (4):

$$CIOU = IoU - \left(\frac{\rho^2(b, b^{gt})}{c^2} \right) + \alpha v \quad (1)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (2)$$

$$\alpha = \frac{v}{(1 - IoU) + v} \quad (3)$$

$$Loss_{IoU} = 1 - IoU + \left(\frac{\rho^2(b, b^{gt})}{c^2} \right) + \alpha v \quad (4)$$

In the above equation, IoU represents the intersection over union between the predicted bounding box and the ground truth bounding box. $\rho^2(b, b^{gt})$ indicating the Euclidean distance of center point b and the center point of the ground truth bounding box b^{gt} ; c is the diagonal length of the smallest closed region enclosing both the predicted bounding box and the ground truth bounding box. $\rho^2(b, b^{gt})/c^2$ is the distance penalty term; v measures the consistency of aspect ratios between predicted bounding boxes and ground truth boxes [12]; α is the aspect ratio loss coefficient; w^{gt} and h^{gt} are the width and height of the ground truth box; w and h are the width and height of the predicted box.

C. Model Evaluation

In the process of evaluating deep learning models, accuracy, precision, recall and F1 values can be used to evaluate the model.

1) The accuracy rate (Accuracy) is the proportion of the number of samples classified correctly to the total number of samples, and the calculation formula is shown in Formula (5)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

2) Precision represents the proportion of true positives among samples predicted as positive examples, and the calculation formula is shown in Formula(6)

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

3) Recall rate represents the proportion of samples predicted as positive among the true positives, and the calculation formula is shown in Formula (7)

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

4) F1 value (F1 Score) is the harmonic mean of precision and recall, which is used to measure the overall performance of a classifier. When both precision and recall are high, F1 value is also high, as shown in formula (8)

$$F1score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

In summary, accuracy reflects the percentage of correctly predicted results out of the total sample; precision measures how close the average predicted value is to the true value, used to characterize the magnitude of systematic error; recall evaluates the quality of classification results, indicating the proportion of samples correctly classified; F1 score is a harmonic mean of precision and recall, used for comprehensive evaluation. Additionally, IoU (Intersection over Union, Intersection over Union) can be used as an evaluation metric [13], primarily in object detection, assessing localization accuracy by calculating the overlap rate between predicted bounding boxes and real bounding boxes.

$$IoU = \frac{TP}{FP + TP + FN} \quad (9)$$

In the formula: TP, which means that the prediction is 1 and the actual is 1, and the prediction is correct, that is, the true example.

FP indicates that the prediction is 1 and the actual is 0, which is a prediction error, that is, false negative; FN indicates that the prediction is 0 and the actual is 1, which is a prediction error, that is, true negative.

IV. DATA SET CONSTRUCTION AND MODEL TRAINING

A. Data Set Construction

This design employs the Roboflow tool for data annotation, meticulously categorizing images of photovoltaic panels collected from the internet and laboratories into six levels: Almost Clean, Clean, Completely Dirty, Dirty, Slightly Dirty, and Very Dirty. These categories are further scientifically allocated in a 7:2:1 ratio to form the training set, validation set, and test set [14], ensuring an even distribution of samples across each subset.

The pictures of photovoltaic panels with different dirt levels are shown in sequence.

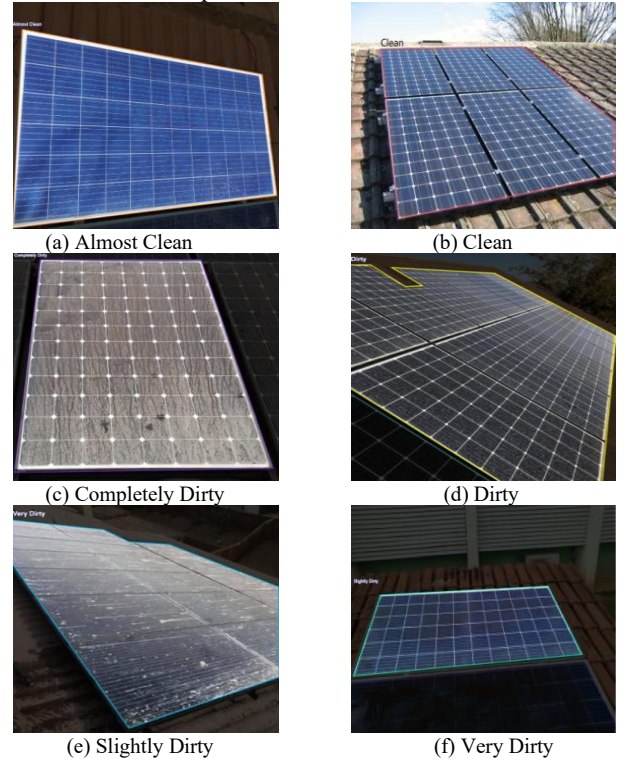


Fig. 3 Photovoltaic panels with varying degrees of soiling

B. Model Training

Experimental environment: In this paper, the training and testing of the model are realized on PC. The experimental environment is Ubuntu 20.04, CUDA12.2, the graphics card is GeForce RTX4060, the network development environment is PyTorch 1.13.1, the development language is Python3.10, and the integrated development environment is Jupyter.

TABLE I
ENVIRONMENT TABLE

Name	Environment parameter
operating system	Ubuntu 20.04
CPU	Intel Core i5-12400F
GPU	Nvidia GeForce RTX 4060
internal storage	16GB
Python	3.10
Pytorch	1.13.1
Cuda	12.2

The training rounds of the experimental setup are 300, the batch size is 16, the initial learning rate is set to 0.001. This allows for effective gradient descent and parameter updates without excessively consuming computational resources. The optimizer is set to Adam, and the IoU threshold is set to 0.5. The ratio of the number of training sets, verification sets and test sets is set to 7:2:1.

Based on the training of the dataset, two models, best.pt and last.pt, were finally generated. The best.pt model is the most accurate one, with strong capability in identifying the degree of dirt on photovoltaic panels, making it suitable for practical applications. To verify whether these models can be applied in future applications, this design also developed a front-end interface based on PyQt5. This interface can perform real-time detection of dirt levels on photovoltaic panels using images, videos, and camera applications, providing a rating, as shown in Figure 4.

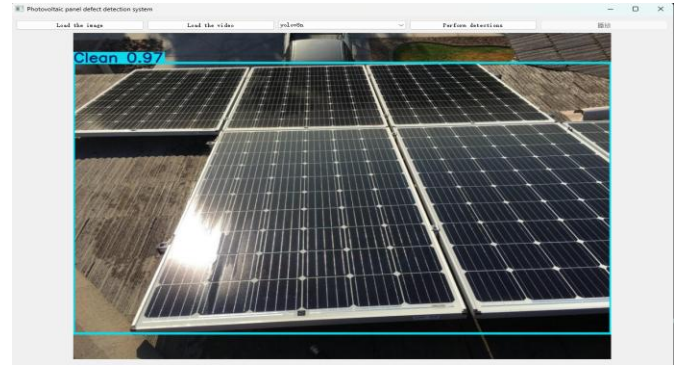


Fig. 4 Terminal interface

V. EXPERIMENTAL ANALYSIS

By configuring the model parameters, the model was trained for 3 hours and 14 minutes, as shown in Figure 5. The horizontal axis represents the number of training epochs (epochs). From the figure, it can be seen that both the training loss curve and the validation loss curve decrease gradually with increasing training iterations. In particular, the rate of decline is fastest in the first 50 epochs, and the gradient becomes smaller over time [15], indicating that the model is gradually improving. The four indicators of accuracy, recall rate, mAP50 and mAP50-95 are opposite to each other. With the increase of training times, they continue to increase and finally tend to be stable, indicating the training data.

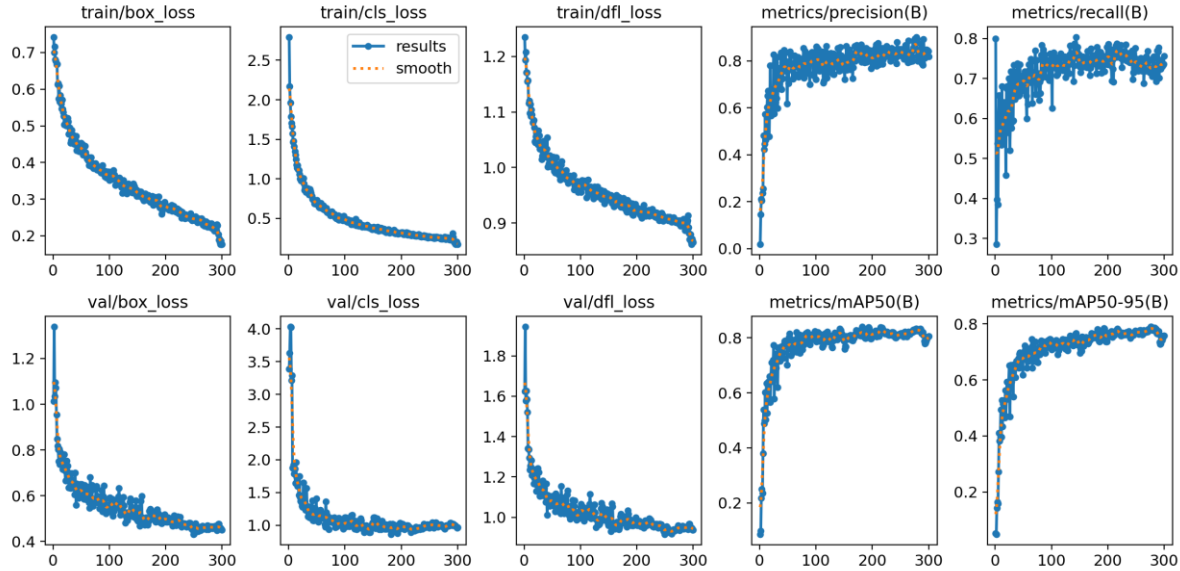
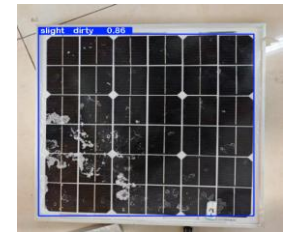


Fig. 5 Result curve

As shown in Figure 6, through a certain amount of model training, it can be observed that the exported model achieves a comprehensive confidence level above 0.9 in practical applications, meeting the requirements for experimental operations and real-world tasks. The model proposed in this paper successfully detected all targets without any missed or false detections, while the recognition speed and accuracy also met the expected conditions.



(a) 0.90 confidence



(b) 0.86 confidence

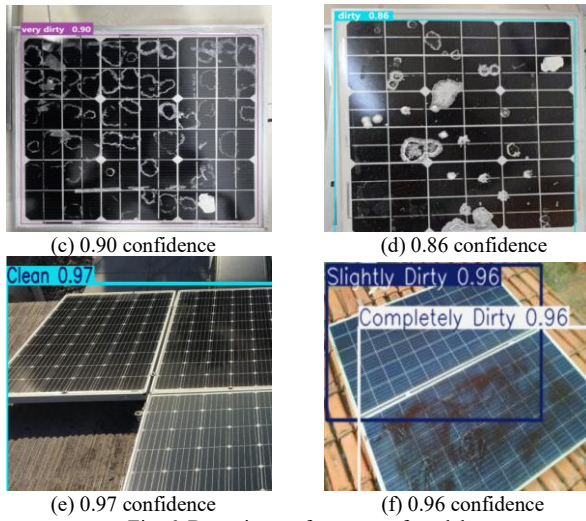


Fig. 6 Detection performance of models

To validate the superiority of the YOLOv8n model, comparative experiments were conducted with other state-of-the-art object detection algorithms, including YOLOv5n, YOLOv7n and Faster R-CNN. The experiments were performed under the same dataset and hardware environment (RTX 4060, Ubuntu 20.04), with evaluation metrics including mAP@0.5, model parameters, and inference speed (FPS).

Table II
Performance Comparison of Different Models

Model	mAP@0.5	Parameters	FPS
YOLOv8n	0.832	3.2	82
YOLOv5n	0.791	3.9	75
YOLOv7n	0.810	5.1	68
Faster R-CNN	0.725	28.5	22

The YOLOv8n model demonstrates significant advantages in accuracy, lightweight design, and real-time performance compared to YOLOv5n, YOLOv7n, and Faster R-CNN. Its balanced performance makes it an ideal choice for photovoltaic panel contamination detection, particularly for edge deployment in cleaning robot systems. The experimental results validate that the proposed system outperforms traditional and lightweight detection models in practical industrial applications.

VI. CONCLUSIONS

In view of the decrease in power generation efficiency (5%-30%) caused by photovoltaic panel dirt and the limitations of traditional inspection methods, an intelligent identification system based on YOLOv8n is proposed to realize real-time detection of dirt degree and six levels of classification (Almost Clean, Clean, Completely Dirty, Dirty, Slightly Dirty, Very Dirty), so as to provide data support for the scheduling of cleaning robots.

The intelligent recognition system for YOLOv8n photovoltaic panel contamination constructed in this paper effectively addresses the issue of quantifying contamination levels, achieving a balance between lightweight design and detection accuracy (mAP@0.5=0.832). This provides an efficient solution for the automation of photovoltaic

operations. The system supports edge deployment and real-time interaction, significantly enhancing the scheduling efficiency of cleaning robots and effectively reducing equipment damage, thereby contributing to the efficient and stable operation of photovoltaic systems. Future improvements can focus on enhancing robustness in complex environments and promoting deeper collaboration with for intelligent operation and maintenance.

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